

File: saas_subscription_churn
Rows: 147,896 | Fields: 52

Key Metrics

Metric	Value
Churn Rate	9.9%
MRR	329563958
ARR	23880
Support Tickets	2000
Avg Satisfaction	3.98

Dataset Overview

- File: saas_subscription_churn
- Records: 147,896 | Fields: 52
- Missing values: Yes
- Duplicate rows: 0

Fields

subscription_id (object), account_id (object), start_date (object), end_date (object), plan_tier_ravenstack_subscriptions (object), seats_ravenstack_subscriptions (int64), mrr_amount (int64), arr_amount (int64), is_trial_ravenstack_subscriptions (bool), upgrade_flag (bool), downgrade_flag (bool), churn_flag_ravenstack_subscriptions (bool), billing_frequency (object), auto_renew_flag (bool), account_name (object), industry (object), country (object), signup_date (object), referral_source (object), plan_tier_ravenstack_accounts (object)

... and 32 more fields

Field Semantics

Field	Inferred Meaning	Confidence
subscription_id	Unique identifier	high
account_id	Unique identifier	high
start_date	Date field	high
end_date	Date field	high
plan_tier_ravenstack_subscriptions	Unknown — review needed	low
seats_ravenstack_subscriptions	Unknown — review needed	low
mrr_amount	Monthly recurring revenue (MRR)	high
arr_amount	Annualized recurring revenue (ARR)	high
is_trial_ravenstack_subscriptions	Status / boolean flag	medium
upgrade_flag	Upgrade event indicator	medium
downgrade_flag	Downgrade event indicator	medium
churn_flag_ravenstack_subscriptions	Unknown — review needed	low
billing_frequency	Billing cycle frequency (monthly/yearly)	high
auto_renew_flag	Auto-renewal setting flag	medium
account_name	Company / account name	high
industry	Industry / vertical	high
country	Geographic region	high
signup_date	Date field	high
referral_source	Acquisition / source channel	high
plan_tier_ravenstack_accounts	Unknown — review needed	low
seats_ravenstack_accounts	Unknown — review needed	low
is_trial_ravenstack_accounts	Status / boolean flag	medium
churn_flag_ravenstack_accounts	Unknown — review needed	low
usage_id	Unique identifier	high
usage_date	Date field	high

feature_name	Software feature name	high
usage_count	Feature usage count	high
usage_duration_secs	Usage duration in seconds	high
error_count	Error occurrence count	high
is_beta_feature	Status / boolean flag	medium
ticket_id	Unique identifier	high
submitted_at	Ticket submission timestamp	high
closed_at	Ticket closure timestamp	high
resolution_time_hours	Time field	high
priority	Ticket priority level	high
first_response_time_minutes	Time field	high
satisfaction_score	Customer satisfaction score	high
escalation_flag	Escalation indicator	medium
churn_event_id	Unique identifier	high
churn_date	Date field	high
reason_code	Churn reason category code	high
refund_amount_usd	Refund amount in USD	high
preceding_upgrade_flag	Upgrade event preceding churn	medium
preceding_downgrade_flag	Downgrade event preceding churn	medium
is_reactivation	Status / boolean flag	medium
feedback_text	Customer feedback / ticket description	medium
satisfaction	Unknown — review needed	low
churn_flag	Churn status indicator	high
ticket	Unknown — review needed	low
churn	Churn status indicator	high
mrr	Recurring revenue	high
arr	Recurring revenue	high

Data Quality

[WARN] Missing values in end_date

133296 rows (90.1%) missing

Affects: Analysis involving this field will have reduced sample size

[WARN] Missing values in satisfaction_score

62723 rows (42.4%) missing

Affects: Analysis involving this field will have reduced sample size

[WARN] Missing values in churn_event_id

29841 rows (20.2%) missing

Affects: Analysis involving this field will have reduced sample size

[WARN] Missing values in churn_date

29841 rows (20.2%) missing

Affects: Analysis involving this field will have reduced sample size

[WARN] Missing values in reason_code

29841 rows (20.2%) missing

Affects: Analysis involving this field will have reduced sample size

[WARN] Missing values in refund_amount_usd

29841 rows (20.2%) missing

Affects: Analysis involving this field will have reduced sample size

[WARN] Missing values in preceding_upgrade_flag

29841 rows (20.2%) missing

Affects: Analysis involving this field will have reduced sample size

[WARN] Missing values in preceding_downgrade_flag

29841 rows (20.2%) missing

Affects: Analysis involving this field will have reduced sample size

[WARN] Missing values in is_reactivation

29841 rows (20.2%) missing

Affects: Analysis involving this field will have reduced sample size

[WARN] Missing values in feedback_text

58601 rows (39.6%) missing

Affects: Analysis involving this field will have reduced sample size

[WARN] Missing values in satisfaction

62723 rows (42.4%) missing

Affects: Analysis involving this field will have reduced sample size

Key Findings

seats_ravenstack_subscriptions Top/Bottom by plan_tier_ravenstack_subscriptions

Evidence: Top: Pro (1,481,774.00), Bottom: Basic (1,374,201.00). Range ratio: 1.1x

Interpretation: plan_tier_ravenstack_subscriptions is a strong differentiator for seats_ravenstack_subscriptions. The gap between top and bottom segments suggests significant optimization opportunity.

Action: Investigate what drives success in top plan_tier_ravenstack_subscriptions segments and apply best practices to lower performers.

mrr_amount Top/Bottom by plan_tier_ravenstack_subscriptions

Evidence: Top: Enterprise (246,409,760.00), Bottom: Basic (22,520,814.00). Range ratio: 10.9x

Interpretation: plan_tier_ravenstack_subscriptions is a strong differentiator for mrr_amount. The gap between top and bottom segments suggests significant optimization opportunity.

Action: Investigate what drives success in top plan_tier_ravenstack_subscriptions segments and apply best practices to lower performers.

seats_ravenstack_subscriptions Top/Bottom by industry

Evidence: Top: FinTech (1,026,950.00), Bottom: EdTech (775,791.00). Range ratio: 1.3x

Interpretation: industry is a strong differentiator for seats_ravenstack_subscriptions. The gap between top and bottom segments suggests significant optimization opportunity.

Action: Investigate what drives success in top industry segments and apply best practices to lower performers.

mrr_amount Top/Bottom by industry

Evidence: Top: FinTech (85,670,242.00), Bottom: EdTech (54,467,247.00). Range ratio: 1.6x

Interpretation: industry is a strong differentiator for mrr_amount. The gap between top and bottom segments suggests significant optimization opportunity.

Action: Investigate what drives success in top industry segments and apply best practices to lower performers.

Thinking Models

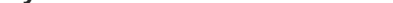
1. ■■■■ (Decomposition)

Evidence: plan_tier_ravenstack_subscriptions, billing_frequency, industry

Takeaway:

2. ■■■■ (Segment Divergence)

[illegible]

Takeaway: 

3. ■■■■ (Proxy Inference)

Evidence: ■ 14

Takeaway:

4. ■■■■■ (Leverage Point)

Evidence: ██████████/██████████"████"████████████████

Takeaway: ■■■■■ + ■■■■■■■■■■■■■■■■■■

5. ■■ vs ■■ (Constraint vs Preference)

[illegible][illegible]

Metrics Glossary

Metric	Value	Formula
Churn Rate	9.9%	$SUM(churn_flag) / COUNT(*) = 14600 / 147896 = 9.9\%$
MRR	329563958	$SUM(mrr_amount) = 329,563,958.00$
ARR	23880	$Latest\ arr_amount = 23,880.00$
Support Tickets	2000	$COUNT(ticket_id) = 2,000$
Avg Satisfaction	3.98	$AVG(satisfaction_score) = 3.98$

Data Tables

■■■■■

■■	■■	■■	■■■
ravenstack_accounts	500	10	account_id, account_name
ravenstack_churn_events	600	9	churn_event_id, account_id
ravenstack_feature_usage	25000	8	usage_id, subscription_id
ravenstack_subscriptions	5000	14	subscription_id, account_id
ravenstack_support_tickets	2000	9	ticket_id, account_id, closed_at

■plan_tier_ravenstack_subscriptions■seats_ravenstack_subscriptions

plan_tier_ravenstack_subscriptions	seats_ravenstack_subscriptions_■■	seats_ravenstack_subscriptions_■	■■■
Pro	1481774	30.09	49252
Enterprise	1471373	29.01	50728
Basic	1374201	28.68	47916

■plan_tier_ravenstack_subscriptions■mrr_amount

plan_tier_ravenstack_subscriptions	mrr_amount_■■	mrr_amount_■■	■■■
Enterprise	246409760	4857.47	50728
Pro	60633384	1231.08	49252
Basic	22520814	470.01	47916

■billing_frequency■seats_ravenstack_subscriptions

billing_frequency	seats_ravenstack_subscriptions_■■	seats_ravenstack_subscriptions_■	■■■
monthly	2197802	29.38	74799
annual	2129546	29.13	73097

■billing_frequency■mrr_amount

billing_frequency	mrr_amount_■■	mrr_amount_■■	■■■
annual	166936285	2283.76	73097
monthly	162627673	2174.2	74799

■industry■seats_ravenstack_subscriptions

industry	seats_ravenstack_subscriptions_■■	seats_ravenstack_subscriptions_■	■■■
FinTech	1026950	30.41	33772
DevTools	870730	28.11	30979
HealthTech	850336	28.34	30000
Cybersecurity	803541	27.5	29221
EdTech	775791	32.43	23924

■industry■mrr_amount

industry	mrr_amount_■■	mrr_amount_■■	■■■
FinTech	85670242	2536.72	33772

industry	mrr_amount_■■	mrr_amount_■■	■■■
HealthTech	64352050	2145.07	30000
DevTools	62546335	2018.99	30979
Cybersecurity	62528084	2139.83	29221
EdTech	54467247	2276.68	23924

seats_ravenstack_subscriptions■■■

■■■	seats_ravenstack_subscriptions
count	147896.0
mean	29.26
std	22.63
min	1.0
25%	13.0
50%	24.0
75%	39.0
max	189.0

mrr_amount■■■

■■■	mrr_amount
count	147896.0
mean	2228.35
std	3359.36
min	0.0
25%	266.0
50%	882.0
75%	2786.0
max	33830.0

■■■■■

■A	■B	■■■	■■■■	■■■(A/B)	■■
ravenstack_accounts	ravenstack_churn_events	account_id	one_to_many	70.4% / 100.0%	safe
ravenstack_accounts	ravenstack_subscriptions	account_id	one_to_many	100.0% / 100.0%	safe
ravenstack_accounts	ravenstack_subscriptions	plan_tier	many_to_many	100.0% / 100.0%	high
ravenstack_accounts	ravenstack_subscriptions	churn_flag	many_to_many	100.0% / 100.0%	high
ravenstack_accounts	ravenstack_subscriptions	is_trial	many_to_many	100.0% / 100.0%	high
ravenstack_accounts	ravenstack_subscriptions	seats	many_to_one	100.0% / 59.3%	safe
ravenstack_accounts	ravenstack_support_tickets	account_id	one_to_many	98.4% / 100.0%	safe
ravenstack_churn_events	ravenstack_subscriptions	account_id	many_to_one	100.0% / 70.4%	safe
ravenstack_feature_usage	ravenstack_subscriptions	subscription_id	many_to_one	100.0% / 99.3%	safe
ravenstack_subscriptions	ravenstack_support_tickets	account_id	many_to_many	98.4% / 100.0%	moderate
ravenstack_churn_events	ravenstack_support_tickets	account_id	many_to_one	98.6% / 70.5%	safe

■■■■■

■■	■A	■B	■	■■■■	■■
Step 1	ravenstack_subscriptions	ravenstack_accounts	account_id	left	safe
Step 2	ravenstack_subscriptions	ravenstack_feature_usage	subscription_id	left	safe

■ ■	■ A	■ B	■	■ ■ ■ ■	■ ■
Step 3	ravenstack_accounts	ravenstack_support_tickets	account_id	left	safe
Step 4	ravenstack_accounts	ravenstack_churn_events	account_id	left	safe

Recommendations

Top-Bottom

Action:

Metric to Track:

Data Needed:

Action:

Metric to Track:

Data Needed:

Action:

Metric to Track:

Data Needed: